

# GavinKlorfine–Project1\_Supplemental

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## Question 1

**Minnesota High School Graduates:** The Hoyt data in **vcdExtra** gives a  $4 \times 3 \times 7 \times 2$  table classifying nearly 14000 graduates by (a) post-high school Status, (b) Rank in their graduating class, (c) father's Occupational status (7 levels, from 1=High to 7=Low) and (d) Sex. Status is considered the outcome variable. How does it depend on the other factors?

### Part (a)

*Analyze and display the associations between status and each of the other variables individually. In doing this, you might also consider focusing attention on the distinction between College and Non-college Status.*

```
data("Hoyt", package = "vcdExtra")

(status_occup <- (margin.table(Hoyt, c("Status", "Occupation"))) |> CMHtest())
```

```
## Cochran-Mantel-Haenszel Statistics for Status by Occupation
##
##               AltHypothesis  Chisq Df           Prob
## cor             Nonzero correlation  632.26  1 1.6140e-139
## rmeans   Row mean scores differ  644.89  3 1.8705e-139
## cmeans   Col mean scores differ 1381.14  6 2.9297e-295
## general      General association 1479.01 18 1.5446e-303
```

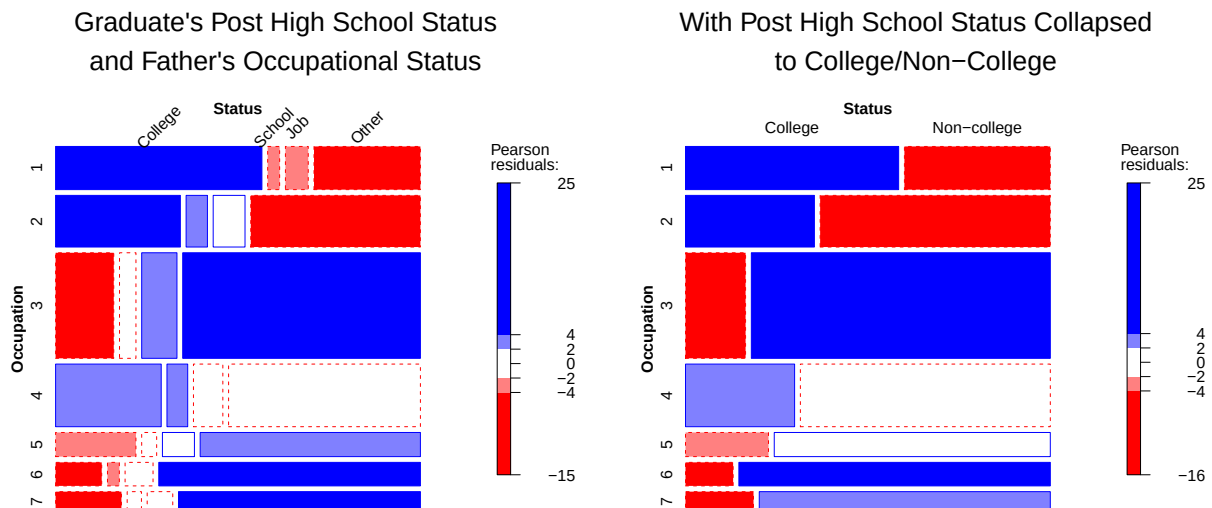
```
# Analyze college vs non-college
distinct_status <- collapse.table(
  Hoyt, Status = c("College", "Non-college", "Non-college", "Non-college")
)

(
  dist_status_occup <- (
    margin.table(distinct_status, c("Status", "Occupation"))) |> CMHtest()
  )
)
```

```
## Cochran-Mantel-Haenszel Statistics for Status by Occupation
##
```

```
##               AltHypothesis  Chisq Df           Prob
## cor             Nonzero correlation  599.63  1 2.0191e-132
## rmeans   Row mean scores differ  599.63  1 2.0191e-132
## cmeans   Col mean scores differ 1392.28  6 1.1353e-297
## general      General association 1392.28  6 1.1353e-297
```

```
mosaic(
  ~ Occupation + Status, data = Hoyt,
  shading = TRUE, gp = shading_Friendly,
  rot_labels = c(top = 45),
  main = "Graduate's Post High School Status\nand Father's Occupational Status"
)
mosaic(
  ~ Occupation + Status, data = distinct_status,
  shading = TRUE, gp = shading_Friendly,
  main = "With Post High School Status Collapsed\nto College/Non-College"
)
```



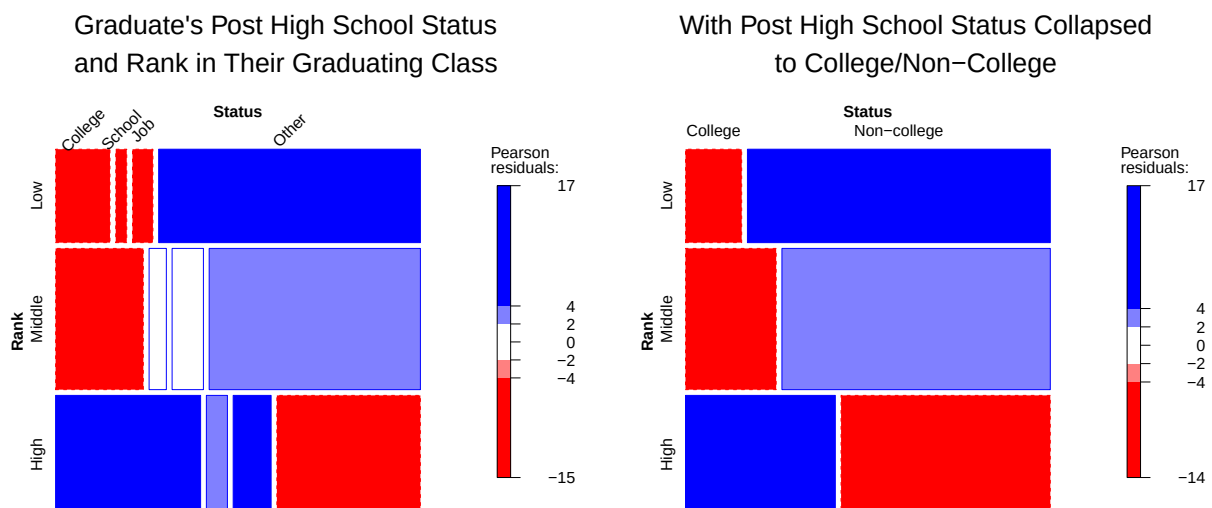
```
(status_rank <- margin.table(Hoyt, c("Status", "Rank")) |> CMHtest())
```

```
## Cochran-Mantel-Haenszel Statistics for Status by Rank
##
##               AltHypothesis  Chisq Df           Prob
## cor             Nonzero correlation  955.79  1 7.3256e-210
## rmeans   Row mean scores differ 1026.23  3 3.6723e-222
## cmeans   Col mean scores differ  971.13  2 1.3208e-211
## general      General association 1045.93  6 1.0393e-222
```

```
(
  dist_status_rank <- (
    margin.table(distinct_status, c("Status", "Rank")) |> CMHtest()
  )
)
```

```
## Cochran-Mantel-Haenszel Statistics for Status by Rank
##
##               AltHypothesis  Chisq Df           Prob
## cor             Nonzero correlation 715.16  1 1.5120e-157
## rmeans   Row mean scores differ 715.16  1 1.5120e-157
## cmeans   Col mean scores differ 734.57  2 3.0848e-160
## general      General association 734.57  2 3.0848e-160
```

```
mosaic(
  ~ Rank + Status, data = Hoyt,
  shading = TRUE, gp = shading_Friendly,
  rot_labels = c(top = 45),
  main = "Graduate's Post High School Status\nand Rank in Their Graduating Class"
)
mosaic(
  ~ Rank + Status, data = distinct_status,
  shading = TRUE, gp = shading_Friendly,
  main = "With Post High School Status Collapsed\nto College/Non-College"
)
```



```
(status_rank <- (margin.table(Hoyt, c("Status", "Sex")) |> assocstats()))
```

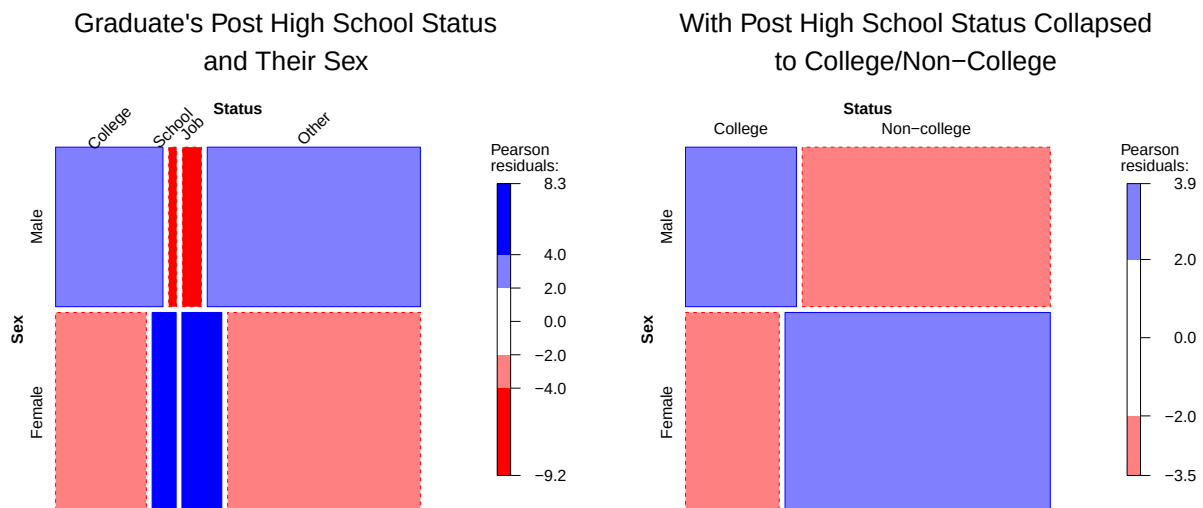
```
##               X^2 df P(> X^2)
## Likelihood Ratio 361.55  3      0
## Pearson          341.87  3      0
##
## Phi-Coefficient   : NA
## Contingency Coeff.: 0.155
## Cramer's V        : 0.156
```

```
(
  dist_status_rank <- (
    margin.table(distinct_status, c("Status", "Sex")) |> assocstats()
  )
)
```

```
)
)

##              X^2 df    P(> X^2)
## Likelihood Ratio 38.816  1 4.6581e-10
## Pearson          38.928  1 4.3980e-10
##
## Phi-Coefficient   : 0.053
## Contingency Coeff.: 0.053
## Cramer's V        : 0.053

mosaic(
  ~ Sex + Status, data = Hoyt,
  shading = TRUE, gp = shading_Friendly,
  rot_labels = c(top = 45),
  main = "Graduate's Post High School Status\nand Their Sex"
)
mosaic(
  ~ Sex + Status, data = distinct_status,
  shading = TRUE, gp = shading_Friendly,
  main = "With Post High School Status Collapsed\ninto College/Non-College"
)
)
```

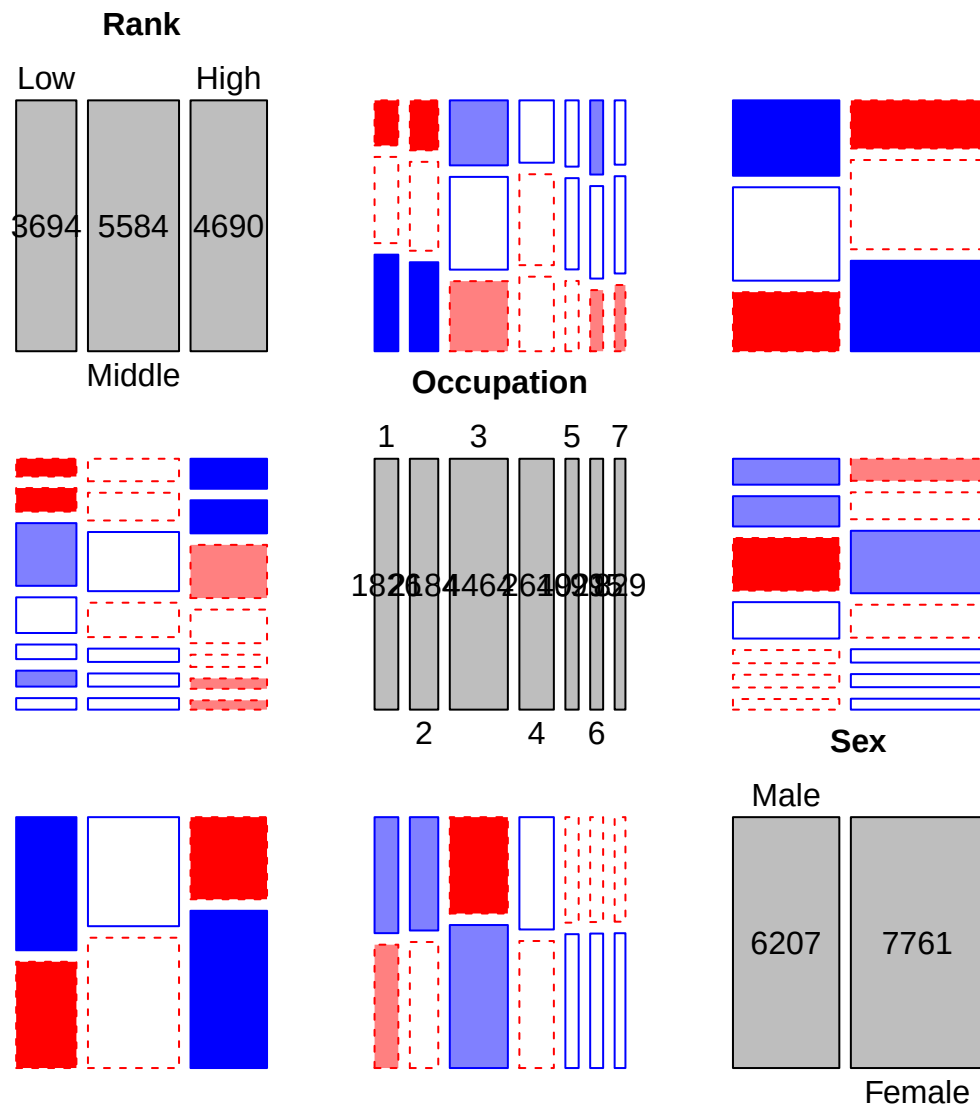


## Part (b)

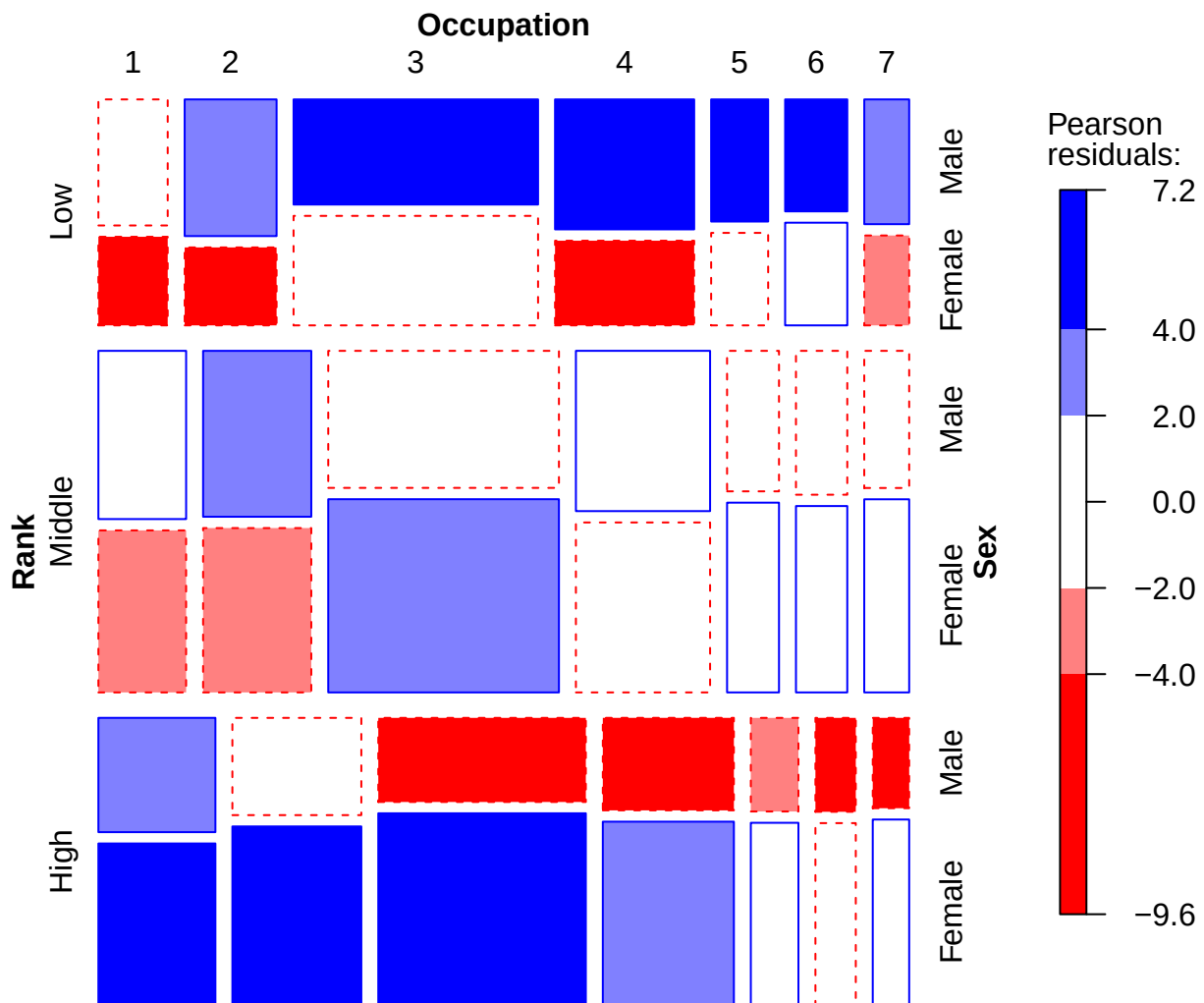
*Consider and analyze the associations among the predictor variables: Rank, Occupation and Sex. Is there anything of interest worth mentioning or analyzing in further detail? Is it useful to consider collapsing some of the categories?*

```
# Display associations
pred_tab <- margin.table(Hoyt, c("Rank", "Occupation", "Sex"))
```

```
pairs(pred_tab, gp = shading_Friendly)
```



```
mosaic(pred_tab, shade = TRUE, gp = shading_Friendly)
```



```
# Collapsing occupation to the following based on similarity in the mosaic plots:
# 1, 2 -> "High",    /    4 -> "Low"
# 3 -> "Middle" /    5, 6, 7 -> "Very Low"

coll.pred_tab <- collapse.table(
  pred_tab,
  Occupation = c("High", "High", "Middle", "Low", "Very Low", "Very Low", "Very Low")
)

# Associations among predictors
CMHtest(margin.table(Hoyt, c("Rank", "Occupation"))) # Test nonzero correlation

## Cochran-Mantel-Haenszel Statistics for Rank by Occupation
##
##           AltHypothesis  Chisq Df    Prob
## cor           Nonzero correlation 108.02  1 2.6609e-25
```

```
## rmeans Row mean scores differ 110.09 2 1.2418e-24
## cmeans Col mean scores differ 167.82 6 1.3026e-33
## general General association 172.06 12 1.8099e-30
```

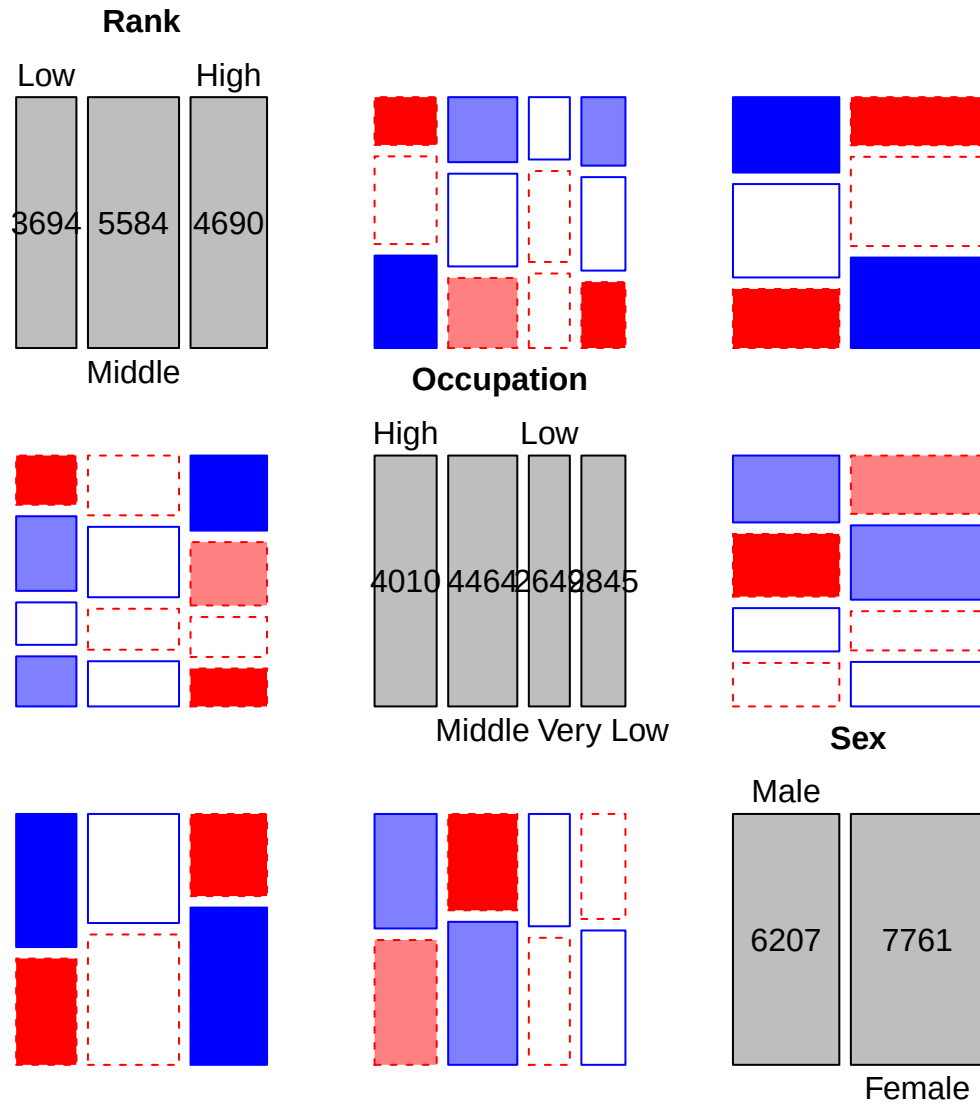
```
CMHtest(margin.table(Hoyt, c("Rank", "Sex"))) # Test difference in column means
```

```
## Cochran-Mantel-Haenszel Statistics for Rank by Sex
##
##           AltHypothesis  Chisq Df       Prob
## cor          Nonzero correlation 383.81  1 1.8469e-85
## rmeans Row mean scores differ 384.30  2 3.5562e-84
## cmeans Col mean scores differ 383.81  1 1.8469e-85
## general General association 384.30  2 3.5562e-84
```

```
CMHtest(margin.table(Hoyt, c("Sex", "Occupation"))) # Test difference in row means
```

```
## Cochran-Mantel-Haenszel Statistics for Sex by Occupation
##
##           AltHypothesis  Chisq Df       Prob
## cor          Nonzero correlation  4.2662  1 3.8877e-02
## rmeans Row mean scores differ  4.2662  1 3.8877e-02
## cmeans Col mean scores differ 58.1352  6 1.0758e-10
## general General association 58.1352  6 1.0758e-10
```

```
pairs(coll.pred_tab, gp = shading_Friendly)
```



### Part (c)

*Start by fitting a simple model of joint independence of Status from the predictors, then add additional associations as you consider necessary until you obtain a reasonably well-fitting model.*

```
coll_Hoyt <- collapse.table(
  Hoyt,
  Occupation = c("High", "High", "Middle", "Low", "Very Low", "Very Low", "Very Low")
)

(joint <- loglm(~Rank*Occupation*Sex + Status, data = coll_Hoyt))
```

## Call:

```
## loglm(formula = ~Rank * Occupation * Sex + Status, data = coll_Hoyt)
##
## Statistics:
##              X^2 df P(> X^2)
## Likelihood Ratio 2520.890 69      0
## Pearson          2512.189 69      0
```

```
(rankstat <- loglm(~Rank*Occupation*Sex + Rank*Status, data = coll_Hoyt))
```

```
## Call:
## loglm(formula = ~Rank * Occupation * Sex + Rank * Status, data = coll_Hoyt)
##
## Statistics:
##              X^2 df P(> X^2)
## Likelihood Ratio 1458.319 63      0
## Pearson          1469.898 63      0
```

```
(occupstat <- loglm(~Rank*Occupation*Sex + Status*(Rank + Occupation), data = coll_Hoyt))
```

```
## Call:
## loglm(formula = ~Rank * Occupation * Sex + Status * (Rank + Occupation),
##       data = coll_Hoyt)
##
## Statistics:
##              X^2 df P(> X^2)
## Likelihood Ratio 430.9455 54      0
## Pearson          408.8948 54      0
```

```
(sexstat <- loglm(~Rank*Occupation*Sex + Status*(Rank + Occupation + Sex), data = coll_Hoyt))
```

```
## Call:
## loglm(formula = ~Rank * Occupation * Sex + Status * (Rank + Occupation +
##       Sex), data = coll_Hoyt)
##
## Statistics:
##              X^2 df      P(> X^2)
## Likelihood Ratio 102.2915 51 2.717215e-05
## Pearson          103.1185 51 2.174678e-05
```

```
(statsexocc <- loglm(~Rank*Occupation*Sex + Status*(Rank + Occupation + Sex) + Status*Sex*Occupation, data = coll_Hoyt))
```

```
## Call:
## loglm(formula = ~Rank * Occupation * Sex + Status * (Rank + Occupation +
##       Sex) + Status * Sex * Occupation, data = coll_Hoyt)
##
## Statistics:
##              X^2 df      P(> X^2)
## Likelihood Ratio 64.11131 42 0.01557704
## Pearson          64.58448 42 0.01411452
```

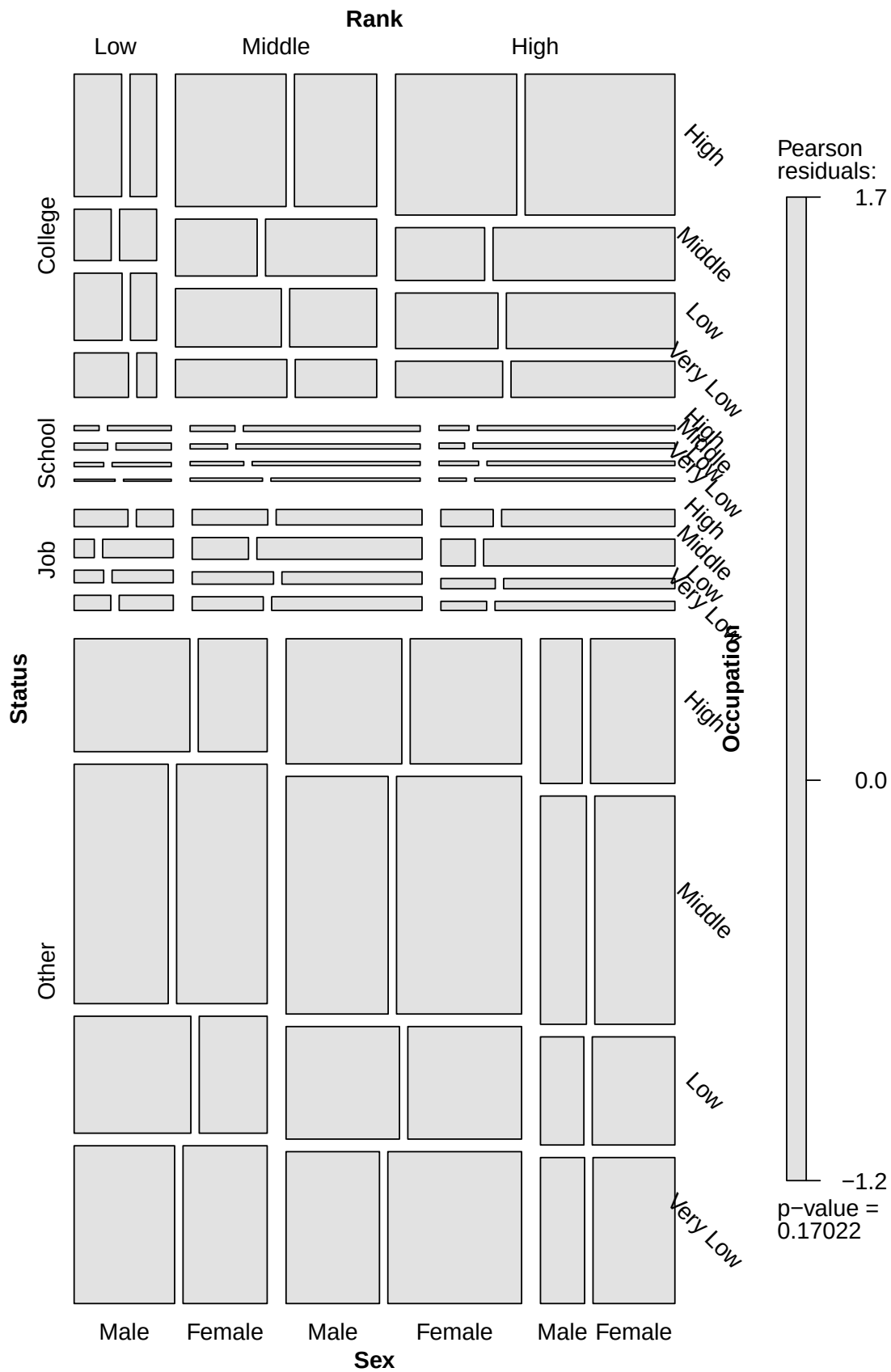
```
(statsexrank <- loglm(~Rank*Occupation*Sex + Status*(Rank + Occupation + Sex) + Status*(Sex*Occupation +
```

```
## Call:
## loglm(formula = ~Rank * Occupation * Sex + Status * (Rank + Occupation +
##      Sex) + Status * (Sex * Occupation + Occupation * Rank), data = coll_Hoyt)
##
## Statistics:
##              X^2 df  P(> X^2)
## Likelihood Ratio 30.41883 24 0.1712554
## Pearson          30.45202 24 0.1702186
```

```
anova(joint, rankstat, occupstat, sexstat, statsexocc, statsexrank)
```

```
## LR tests for hierarchical log-linear models
##
## Model 1:
## ~Rank * Occupation * Sex + Status
## Model 2:
## ~Rank * Occupation * Sex + Rank * Status
## Model 3:
## ~Rank * Occupation * Sex + Status * (Rank + Occupation)
## Model 4:
## ~Rank * Occupation * Sex + Status * (Rank + Occupation + Sex)
## Model 5:
## ~Rank * Occupation * Sex + Status * (Rank + Occupation + Sex) + Status * Sex * Occupation
## Model 6:
## ~Rank * Occupation * Sex + Status * (Rank + Occupation + Sex) + Status * (Sex * Occupation + Occupa
##
##              Deviance df Delta(Dev) Delta(df) P(> Delta(Dev))
## Model 1      2520.89039 69
## Model 2      1458.31891 63 1062.57148         6         0.00000
## Model 3       430.94550 54 1027.37341         9         0.00000
## Model 4       102.29148 51  328.65402         3         0.00000
## Model 5        64.11131 42   38.18017         9         0.00002
## Model 6        30.41883 24   33.69248        18         0.01374
## Saturated      0.00000  0   30.41883        24         0.17126
```

```
mosaic(statsexrank, rot_labels = c(right = -45))
```



## Part (g)

*Conduct a multiple correspondence analysis for the Hoyt data. Provide a reasonable interpretation of the two dimensions that explain the largest proportion of variance.*

```
mjca.coll_Hoyt <- mjca(coll_Hoyt, lambda = "Burt")
summary(mjca.coll_Hoyt)
```

```
##
## Principal inertias (eigenvalues):
##
## dim    value      %   cum%   scree plot
## 1      0.132576  22.4  22.4   *****
## 2      0.095616  16.2  38.6   ****
## 3      0.065541  11.1  49.7   ***
## 4      0.062538  10.6  60.3   ***
## 5      0.061990  10.5  70.8   ***
## 6      0.060105  10.2  81.0   ***
## 7      0.048587   8.2  89.2   **
## 8      0.037438   6.3  95.5   **
## 9      0.026417   4.5 100.0   *
##
## -----
## Total: 0.590807 100.0
##
##
## Columns:
##
##          name    mass  qlt  inr    k=1 cor ctr    k=2 cor ctr
## 1 |      Status:College |   71 779  86 |  652 590 226 |  369 189 101 |
## 2 |      Status:Job |   22 272  98 |  235  21   9 | -812 251 152 |
## 3 |      Status:Other |  145 721  50 | -384 719 162 |   19   2   1 |
## 4 |      Status:School |   12 194 102 |  375  28  13 | -908 166 105 |
## 5 |      Rank:High |   84 539  77 |  520 500 171 | -146  39  19 |
## 6 |      Rank:Low |   66 450  83 | -523 369 137 |  246  81  42 |
## 7 |      Rank:Middle |  100  26  64 |  -90  22   6 |  -40   4   2 |
## 8 |      Occupation:High |   72 570  82 |  532 421 153 |  316 149  75 |
## 9 |      Occupation:Low |   47  10  86 |   28   1   0 |  100   9   5 |
## 10 | Occupation:Middle |   80 321  75 | -279 141  47 | -315 180  83 |
## 11 | Occupation:Very Low |   51 117  86 | -338 115  44 |  -44   2   1 |
## 12 |      Sex:Female |  139 667  50 |  117  65  14 | -357 602 185 |
## 13 |      Sex:Male |  111 667  62 | -146  65  18 |  446 602 231 |
```

```
### BELOW ADAPTED FROM Friendly & Meyer (2016; p. 245), Discrete Data Analysis with R ###
res <- plot(mjca.coll_Hoyt, labels= 0, pch= ".", cex.lab= 1.2)
```

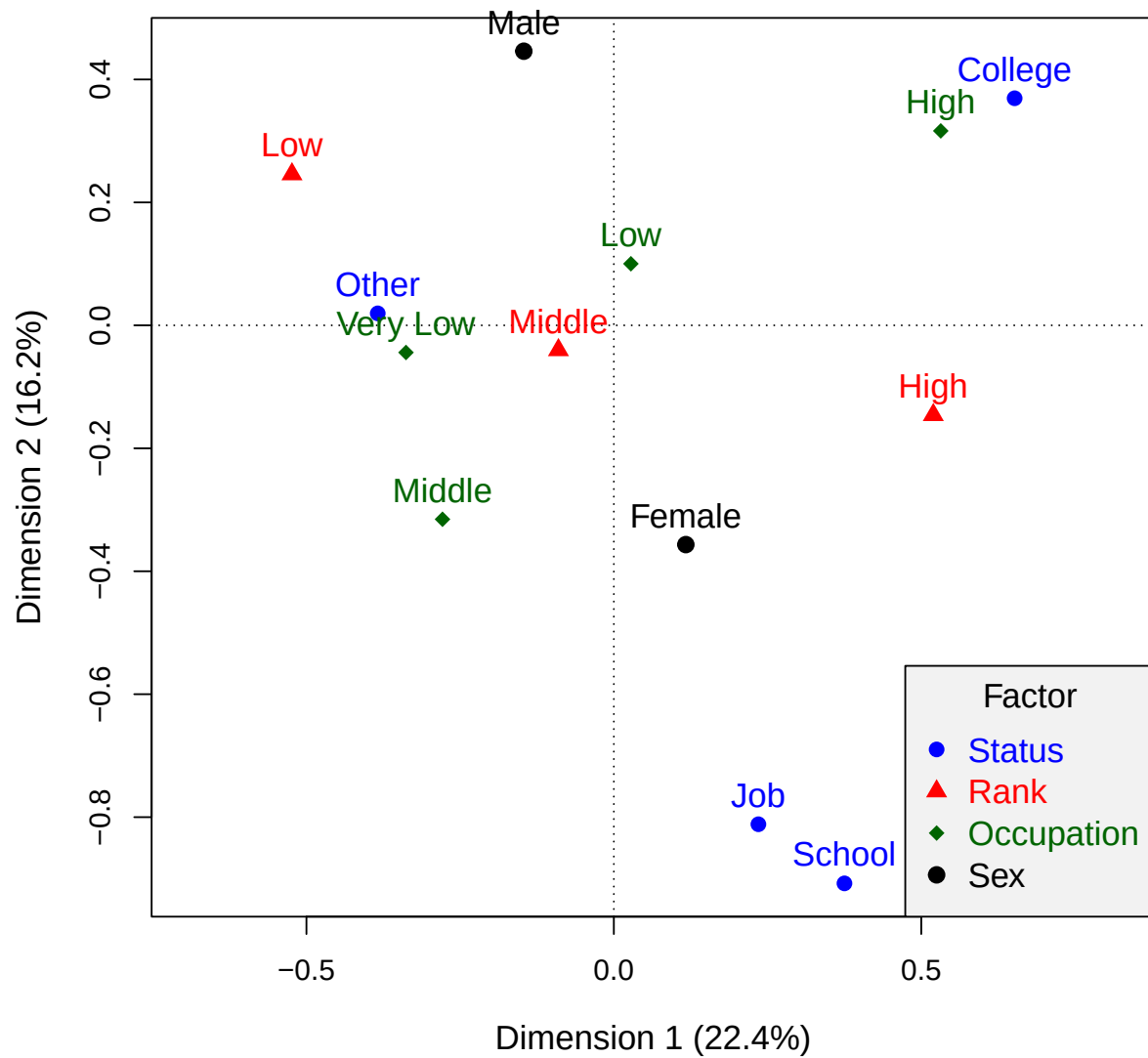
```
# extract factor names and levels
coords <- data.frame(res$cols, mjca.coll_Hoyt$factors)
nlev <- mjca.coll_Hoyt$levels.n
fact <- unique(as.character(coords$factor))

cols <- c("blue", "red", "darkgreen", "black")
points(coords[,1:2], pch=rep(16:19, nlev), col=rep(cols, nlev), cex=1.2)
text(coords[,1:2], label=coords$level, col=rep(cols, nlev), pos=3, cex=1.2, xpd=TRUE)
```

```

legend(
  "bottomright", legend= c("Status", "Rank", "Occupation", "Sex"), title= "Factor",
  title.col= "black", col = cols, text.col = cols, pch= 16:19, bg= "gray95", cex = 1.2
)

```



## Question 2

Student Opinion about the Vietnam War: The data set `Vietnam` in `vcdExtra` gives a  $2 \times 5 \times 4$  contingency table in frequency form reflecting a survey of student opinion on the Vietnam War at the University of North Carolina in May 1967. The table variables are `sex`, `year` in school and `response`, which has categories: (A) Defeat North Vietnam by widespread bombing and land invasion; (B) Maintain the present policy; (C) De-escalate military activity, stop bombing and begin negotiations; (D) Withdraw military forces Immediately. How does the chosen response vary with `sex` and `year`?

### Part (a)

Construct some useful plots showing the proportions of each response category by `sex` and `year` of study. It may be helpful to add some smoothed or fitted curves to aid interpretation.

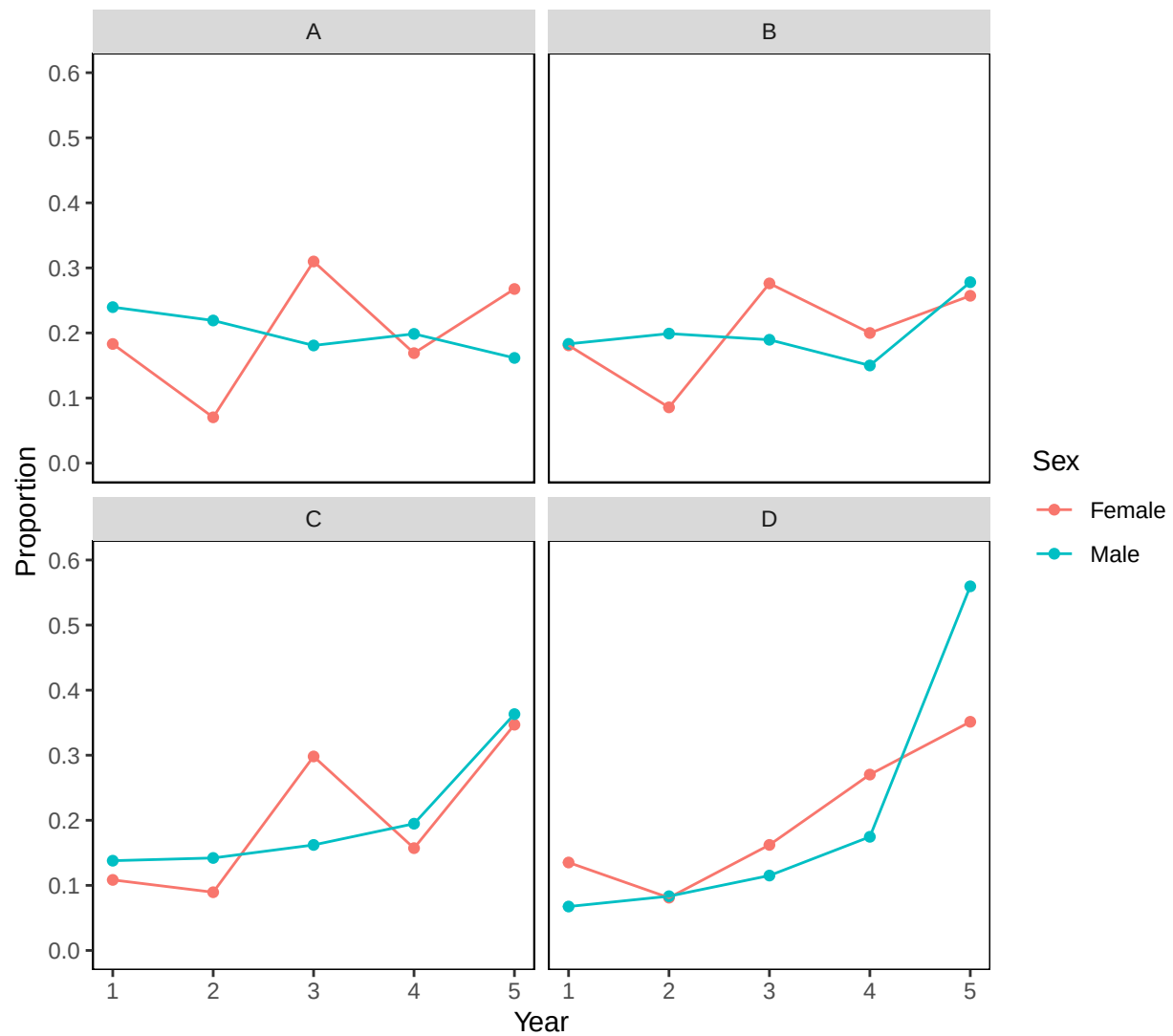
```
data("Vietnam", package = "vcdExtra")

viet <- rename(Vietnam, "Sex" = sex, "Year" = year, "Response" = response)

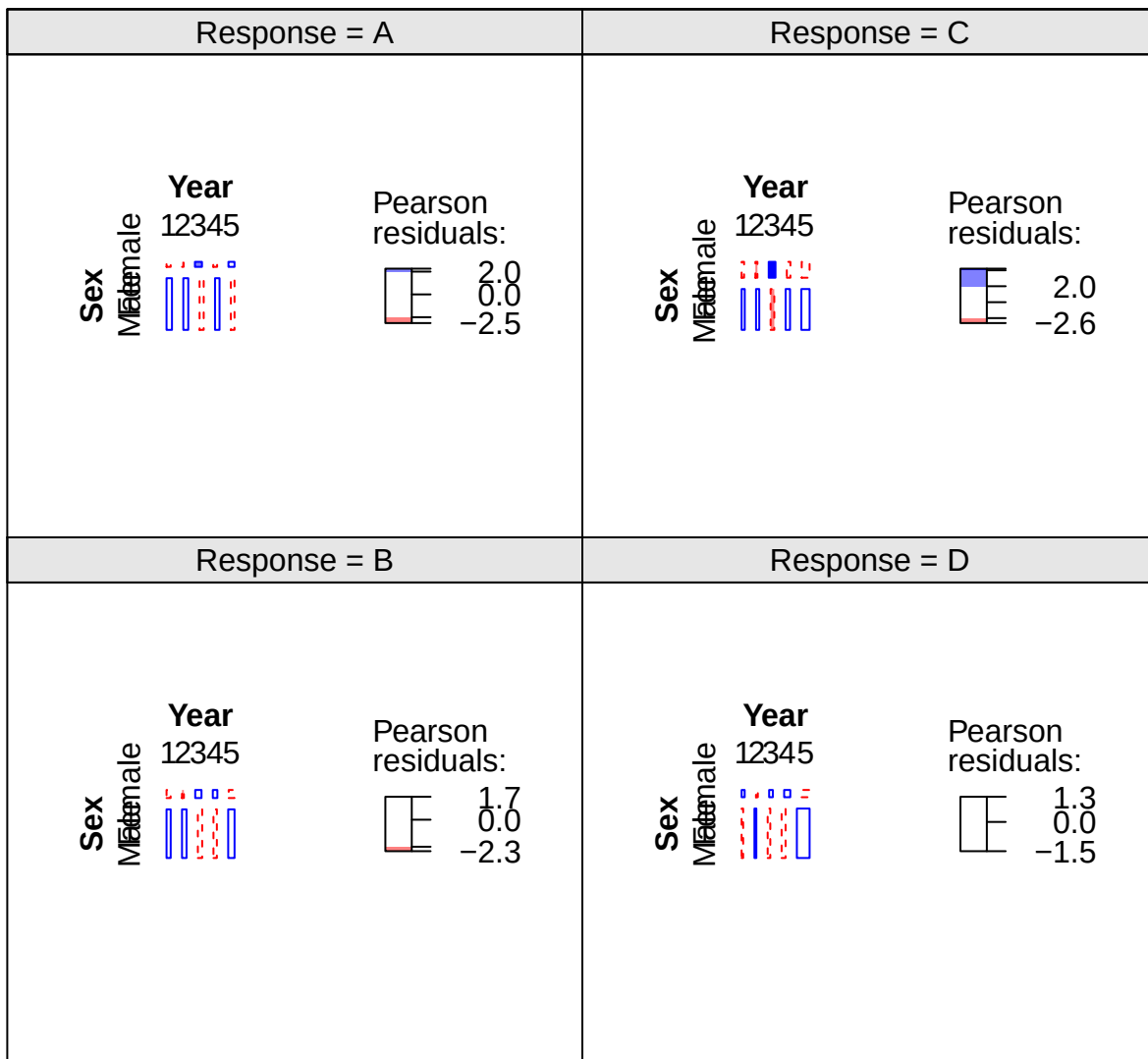
# Male proportions sum to 1 and female proportions sum to 1 within each response category
respProps_viet <- (
  viet |>
  group_by(Response, Sex) |>
  mutate(Proportion = Freq / sum(Freq))
)

ggplot(respProps_viet, aes(x = Year, y = Proportion, fill = Sex, colour = Sex)) +
  facet_wrap(vars(Response)) +
  geom_point() +
  geom_line() +
  scale_y_continuous(breaks = seq(0, 0.6, 0.1)) +
  coord_cartesian(ylim = c(0, 0.6)) +
  labs(
    title = "Proportions of Each Response Category by Sex and Year"
  ) +
  theme(
    plot.title = element_text(hjust = 0.5),
    panel.background = element_blank(),
    panel.border = element_rect()
  )
```

Proportions of Each Response Category by Sex and Year



```
viet_tab <- as_table(viet, freq = "Freq") # See Appendix for as_table()
cotabplot(
  ~ Sex + Year | Response, data = viet_tab,
  shade = TRUE, gp = shading_Friendly,
)
```



## Part (b)

Analyze the associations between response and each of sex and year separately, using graphs and statistical tests. Note that both year and response can be considered ordinal factors.

```
# Create tables
tab.viet_sex <- margin.table(viet_tab, margin = c("Sex", "Response"))
tab.viet_year <- margin.table(viet_tab, margin = c("Year", "Response"))

# Statistical tests for association
CMHtest(tab.viet_sex) # Test row means
```

```
## Cochran-Mantel-Haenszel Statistics for Sex by Response
##
```

```
##               AltHypothesis  Chisq Df    Prob
## cor           Nonzero correlation  65.684  1 5.2947e-16
## rmeans Row mean scores differ  65.684  1 5.2947e-16
## cmeans Col mean scores differ 143.026  3 8.4142e-31
## general      General association 143.026  3 8.4142e-31
```

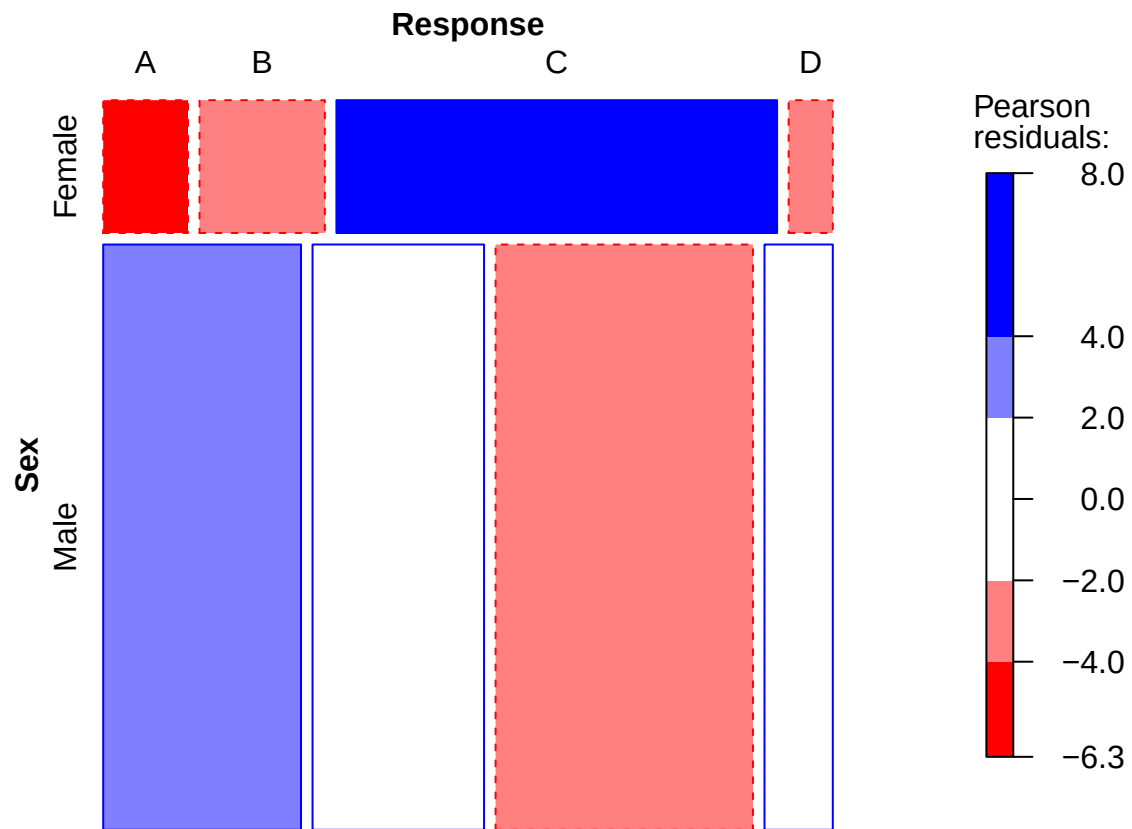
```
CMHtest(tab.viet_year) # Test nonzero correlation
```

```
## Cochran-Mantel-Haenszel Statistics for Year by Response
```

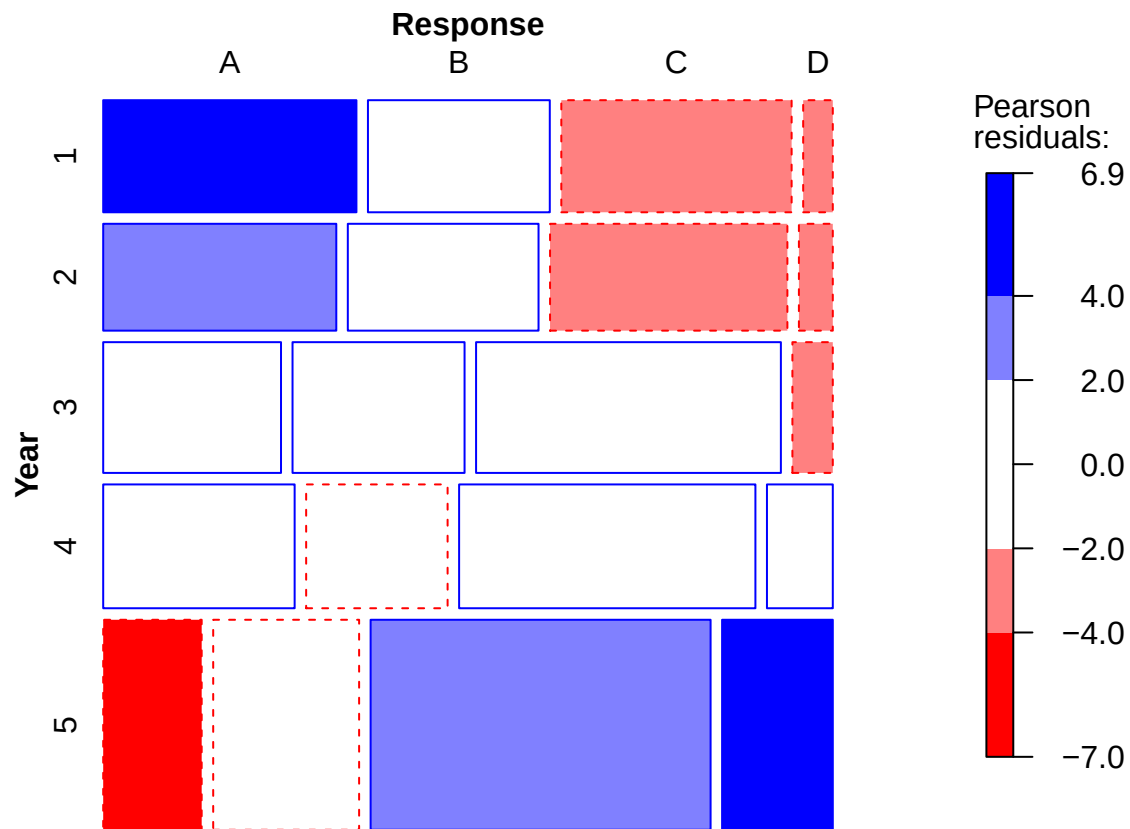
```
##
##               AltHypothesis  Chisq Df    Prob
## cor           Nonzero correlation 174.31  1 8.4543e-40
## rmeans Row mean scores differ 188.84  4 9.3892e-40
## cmeans Col mean scores differ 177.36  3 3.2709e-38
## general      General association 203.97 12 4.9454e-37
```

```
# Graphically analyze associations
```

```
mosaic(tab.viet_sex, shade = TRUE, gp = shading_Friendly)
```



```
mosaic(tab.viet_year, shade = TRUE, gp = shading_Friendly)
```



### Part (c)

*Fit loglinear and/or polytomous logit models designed explain the relationship between response and year of study*

```
viet$Response <- ordered(viet$Response) # Order by "level of pacifism": A < B < C < D

polr.viet_intrcpt <- polr(Response ~ 1, data = viet, Hess = TRUE)
polr.viet <- polr(Response ~ Year, data = viet, Hess = TRUE)

summary(polr.viet)
```

```
## Call:
## polr(formula = Response ~ Year, data = viet, Hess = TRUE)
```

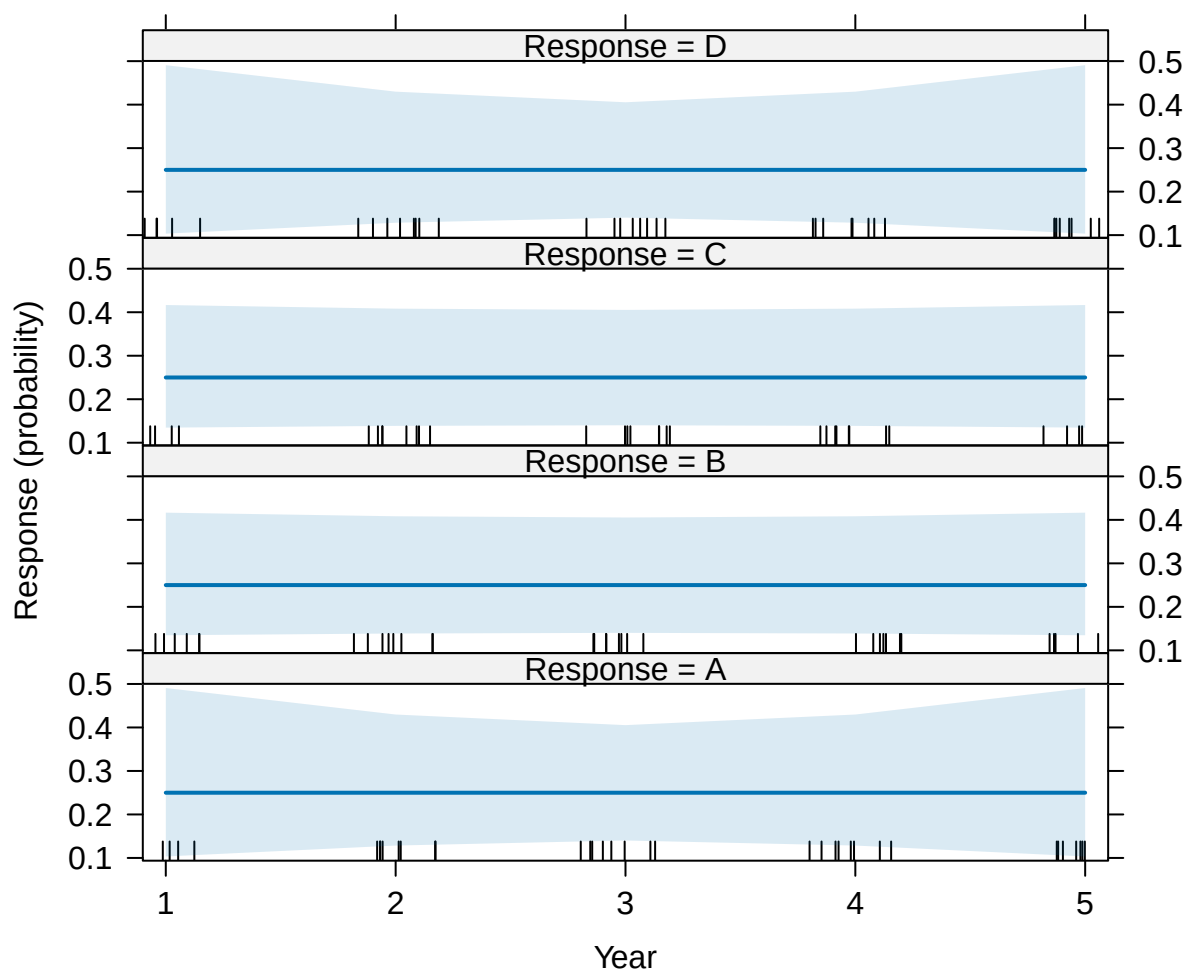
```
##
## Coefficients:
##      Value Std. Error t value
## Year 4.219e-16      0.2 2.11e-15
##
## Intercepts:
##      Value Std. Error t value
## A|B -1.0986  0.7024   -1.5641
## B|C  0.0000  0.6782    0.0000
## C|D  1.0986  0.7024    1.5641
##
## Residual Deviance: 110.9035
## AIC: 118.9035

anova(polr.viet_intrcpt, polr.viet) # Year not significant predictor over and above intercept only model

## Likelihood ratio tests of ordinal regression models
##
## Response: Response
##      Model Resid. df Resid. Dev   Test      Df LR stat. Pr(Chi)
## 1      1      37   110.9035
## 2 Year      36   110.9035 1 vs 2      1      0      1

plot(Effect("Year", polr.viet), main = "Effect Plot for the Effect of Year of Study on Response")
```

## Effect Plot for the Effect of Year of Study on Response

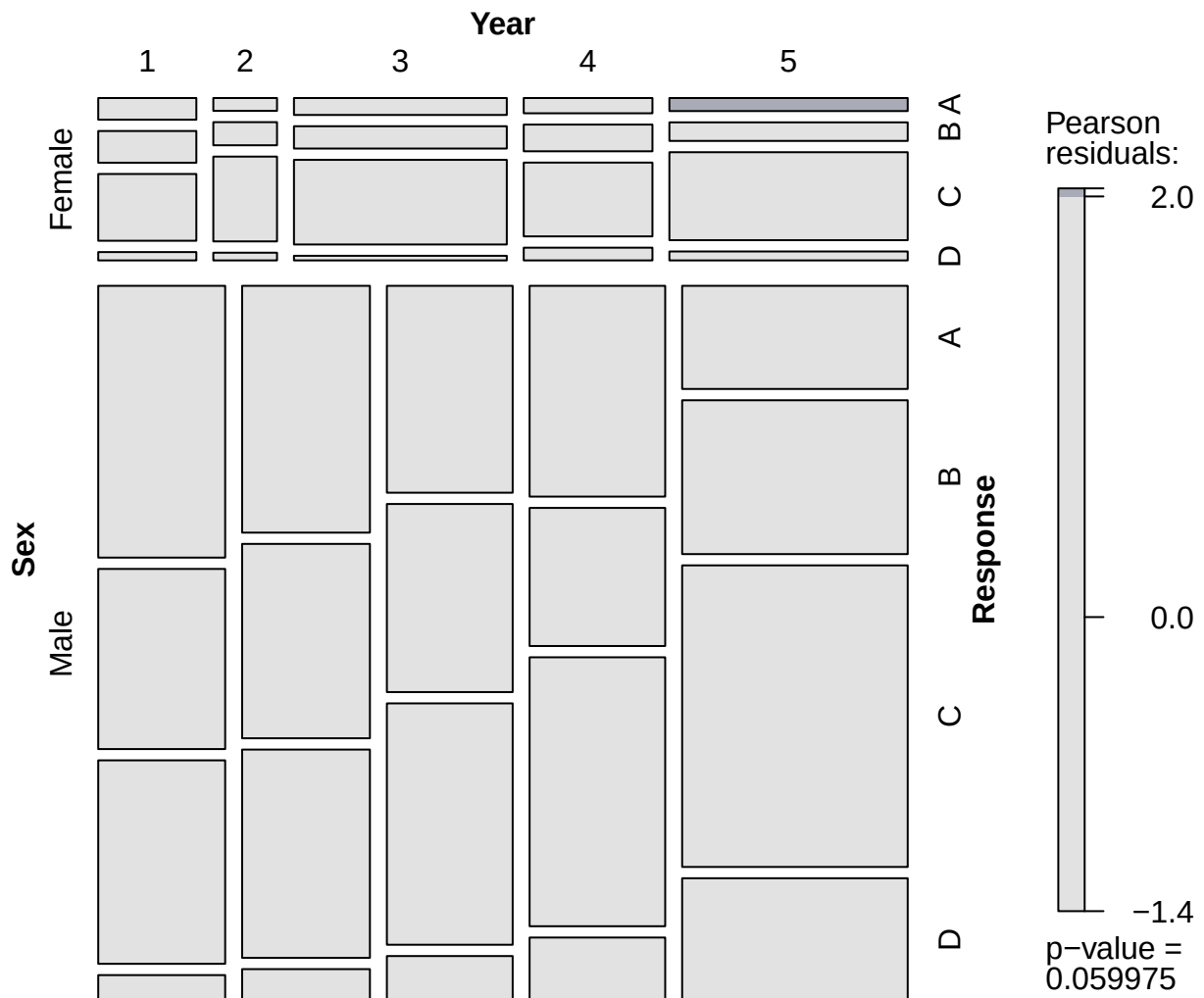


```
loglm.viet <- loglm(~(Year*Sex) + Response*(Year + Sex), data = viet_tab)
summary(loglm.viet)
```

```
## Formula:
## ~(Year * Sex) + Response * (Year + Sex)
## attr("variables")
## list(Year, Sex, Response)
## attr("factors")
##      Year Sex Response Year:Sex Year:Response Sex:Response
## Year      1  0        0       1         1           0
## Sex       0  1        0       1         0           1
## Response  0  0        1       0         1           1
## attr("term.labels")
## [1] "Year"      "Sex"      "Response"  "Year:Sex"
## [5] "Year:Response" "Sex:Response"
```

```
## attr("order")
## [1] 1 1 1 2 2 2
## attr("intercept")
## [1] 1
## attr("response")
## [1] 0
## attr(".Environment")
## <environment: R_GlobalEnv>
##
## Statistics:
##               X^2 df    P(> X^2)
## Likelihood Ratio 19.19418 12 0.08394884
## Pearson          20.39487 12 0.05997543
```

```
mosaic(loglm.viet)
```



## Part (g)

*Conduct a multiple correspondence analysis for the Vietnam data. Provide a reasonable interpretation of the two dimensions that explain the largest proportion of variance.*

```
mjca.viet_tab <- mjca(viet_tab, lambda = "Burt")
summary(mjca.viet_tab)
```

```
##
## Principal inertias (eigenvalues):
##
## dim    value      %   cum%   scree plot
## 1      0.198312  21.6  21.6   *****
## 2      0.152769  16.6  38.3   ****
## 3      0.119945  13.1  51.3   ***
## 4      0.111515  12.2  63.5   ***
## 5      0.103339  11.3  74.7   ***
## 6      0.100824  11.0  85.7   ***
## 7      0.072637   7.9  93.6   **
## 8      0.058425   6.4 100.0   **
##
## -----
## Total: 0.917767 100.0
##
##
## Columns:
##
##      name    mass  qlt  inr    k=1 cor ctr    k=2 cor ctr
## 1 | Sex:Female |    62  678 105 |  869 483 235 | -553 196 123 |
## 2 |   Sex:Male |   272  678  24 | -197 483  53 |  126 196  28 |
## 3 |   Year:1 |    55  211 103 | -533 164  78 | -285  47  29 |
## 4 |   Year:2 |    52  205 104 | -614 205  99 |  -28   0   0 |
## 5 |   Year:3 |    64  353 100 |  276  53  24 | -656 300 179 |
## 6 |   Year:4 |    60   1  99 |  -32   1   0 |   26   0   0 |
## 7 |   Year:5 |   102  645  88 |  444 249 102 |  559 396 210 |
## 8 | Response:A |    85  494  95 | -689 460 203 | -189  35  20 |
## 9 | Response:B |    78   55  93 | -246  55  24 |  -16   0   0 |
## 10 | Response:C |   140  542  74 |  483 477 164 | -178  65  29 |
## 11 | Response:D |    31  593 113 |  332  33  17 | 1379 560 381 |
```

```
### BELOW ADAPTED FROM Friendly & Meyer (2016; p. 245), Discrete Data Analysis with R ###
res <- plot(mjca.viet_tab, labels= 0, pch= ".", cex.lab= 1.2)
```

```
# extract factor names and levels
```

```
coords <- data.frame(res$cols, mjca.viet_tab$factors)
```

```
nlev <- mjca.viet_tab$levels.n
```

```
fact <- unique(as.character(coords$factor))
```

```
cols <- c("blue", "red", "black")
```

```
points(coords[,1:2], pch=rep(16:18, nlev), col=rep(cols, nlev), cex=1.2)
```

```
text(coords[,1:2], label=coords$level, col=rep(cols, nlev), pos=3, cex=1.2, xpd=TRUE)
```

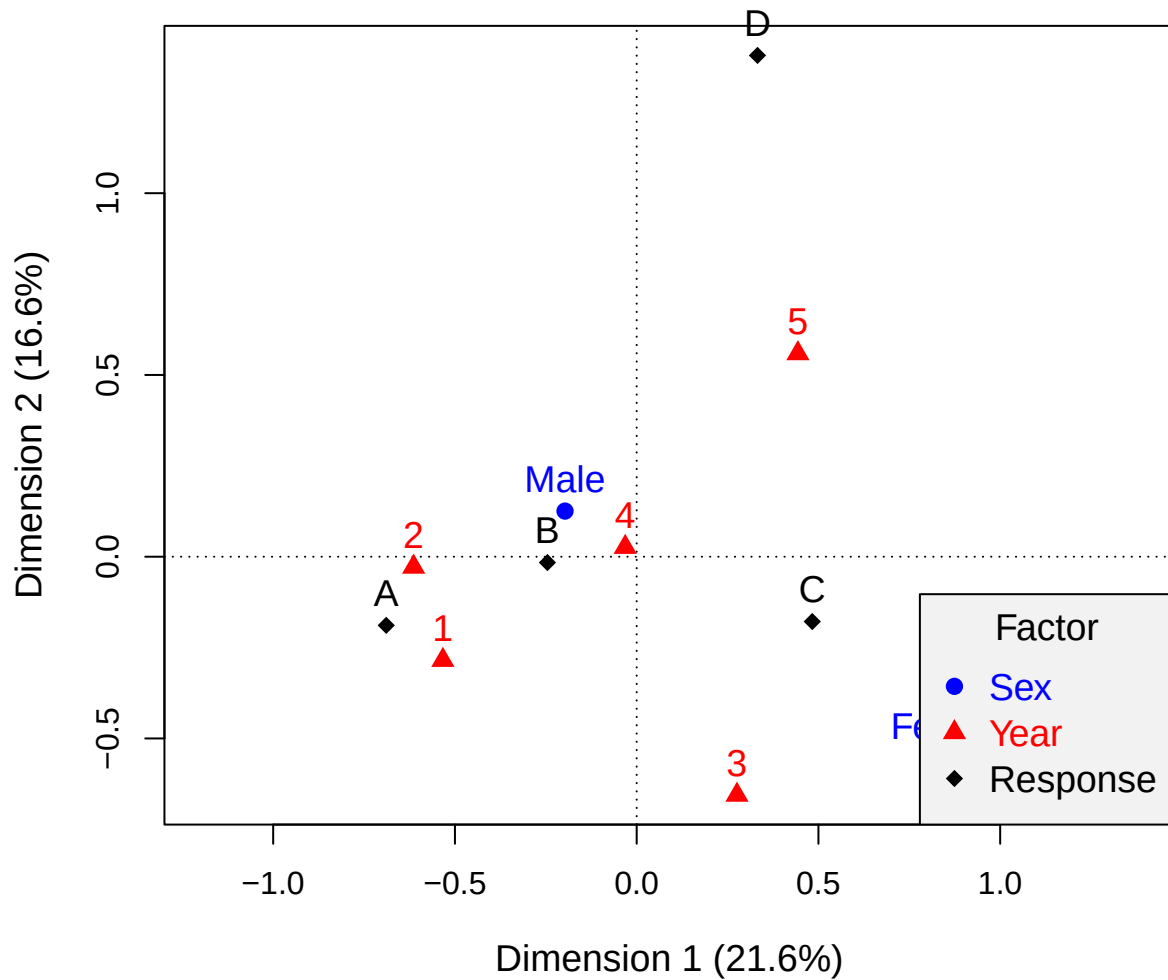
```
legend(
```

```
  "bottomright", legend= c("Sex", "Year", "Response"), title= "Factor",
```

```

title.col= "black", col = cols, text.col = cols, pch= 16:18, bg= "gray95", cex = 1.2
)

```



## Appendix

### as\_table()

Converts object (obj) in frequency or case form into table form. The column containing the frequencies (freq) must be supplied if obj is in frequency form.

```

as_table <- function(obj, freq = NULL, dims = NULL){
  # If user supplied a table or array, return it back to them
  if (length(intersect(c("table", "array"), class(obj))) > 0){

```

```

  tab <- as.table(obj) # Handle arrays

  # To include dimensions if specified
  if (!is.null(dims)){
    tab <- margin.table(tab, margin = dims)
  }

  return(tab)
}

# If obj is a tibble, convert to data frame
else if (length(intersect("tbl", class(obj))) > 0){
  obj <- as.data.frame(obj)
}

if (!is.null(dims)){ # If dims supplied by user, use those
  cols <- dims
}
else { # If dims NOT supplied by user, use everything else
  cols <- colnames(obj)
}

if (!is.null(freq)){ # If freq supplied by user, then... (freq form)
  cols <- cols[cols != freq] # Remove freq column
  tab <- xtabs(reformulate(cols, response = freq), data = obj) # freq ~ cols
}
else { # If freq NOT supplied by user, and not array, then... (case form)
  tab <- xtabs(reformulate(cols), data = obj)

  # Check if user forgot to supply freq, warn if they potentially forgot
  common <- c("n", "freq", "frequency", "count")
  if (length(intersect(tolower(colnames(obj)), common)) > 0){
    warning("Ensure a value for 'freq' was supplied if your data was in frequency form.")
  }
}
return(tab)
}

```